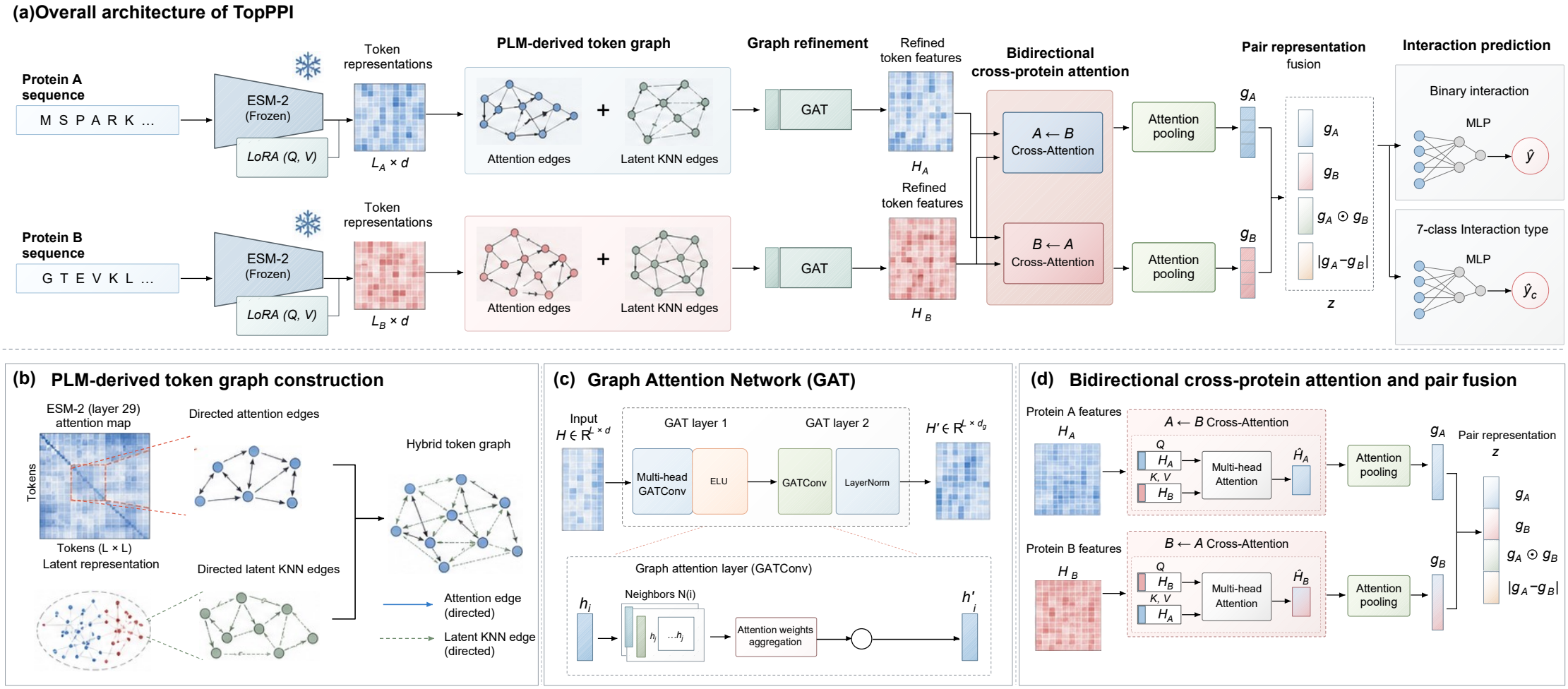

(a)Overall architecture of TopPPI
Protein A sequence
M S P A R K ...
Protein B sequence
G T E V K L ...
ESM-2 (Frozen)
LoRA (Q, V)
Token representations
$L_A \times d$
$L_B \times d$
PLM-derived token graph
Attention edges
Latent KNN edges
Graph refinement
GAT
Refined token features
H_A
H_B
Bidirectional cross-protein attention
$A \leftarrow B$ Cross-Attention
$B \leftarrow A$ Cross-Attention
Attention pooling
g_A
g_B
Pair representation fusion
g_A
g_B
$g_A \odot g_B$
$|g_A - g_B|$
z
Interaction prediction
Binary interaction
MLP
$\hat{y}$
7-class Interaction type
MLP
$\hat{y}_c$
(b) PLM-derived token graph construction
ESM-2 (layer 29) attention map
Tokens
Tokens (L × L)
Latent representation
Directed attention edges
Directed latent KNN edges
Hybrid token graph
Attention edge (directed)
Latent KNN edge (directed)
(c) Graph Attention Network (GAT)
Input $H \in \mathbb{R}^{L \times d}$
GAT layer 1
GAT layer 2
Multi-head GATConv
ELU
GATConv
LayerNorm
$H' \in \mathbb{R}^{L \times d_g}$
Graph attention layer (GATConv)
h_i
Neighbors N(i)
h_j ...h_j
Attention weights aggregation
h'_i
(d) Bidirectional cross-protein attention and pair fusion
Protein A features
H_A
$A \leftarrow B$ Cross-Attention
Q
K, V
Multi-head Attention
$\hat{H}_A$
Attention pooling
g_A
Pair representation
z
g_A
g_B
$g_A \odot g_B$
$|g_A - g_B|$
Protein B features
H_B
$B \leftarrow A$ Cross-Attention
Multi-head Attention
$\hat{H}_B$
Attention pooling
g_B